



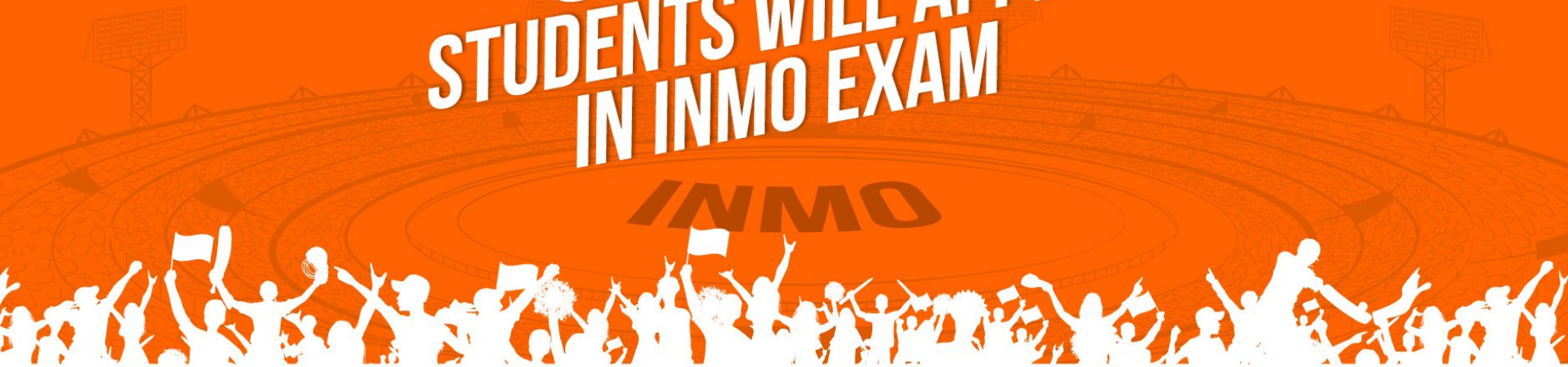
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MATHEMATICS OLYMPIADS BATCH

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BATCH NAME	ELIGIBILITY
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<i>RMO</i>	<i>IOQM Certificates or > 25 marks</i>
<i>INMO</i>	<i>Students who wrote INMO '21</i>

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Kindly attempt the
Session Assignment
that follows



NOTE: Solutions also attached

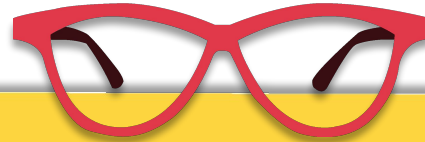
$$(x+a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad -\infty < x < \infty$$

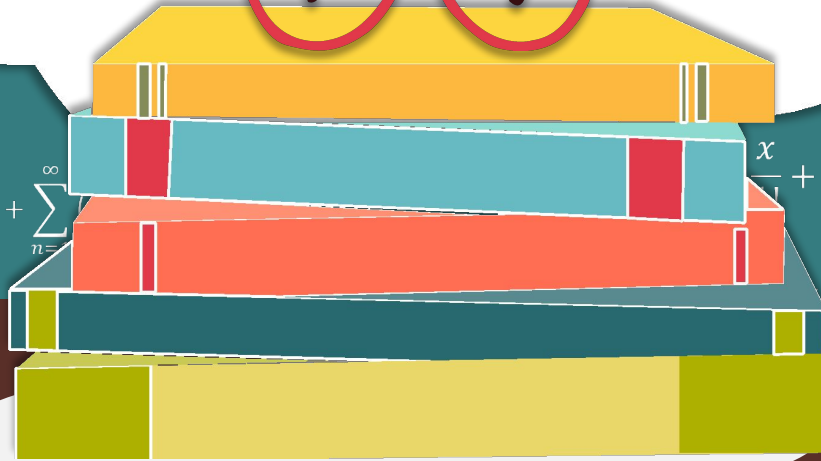
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$(x+a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

Congruence Modulo



$$= a_0 + \sum_{n=1}^{\infty}$$

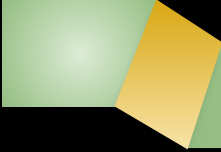


$$(x+a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad -\infty < x < \infty$$

$$\cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad -\infty < x < \infty$$



Congruence

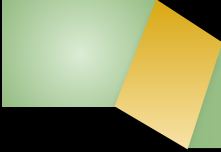
Two integers a and b are said to be **congruent modulo** m ,

i.e. $a \equiv b \pmod{m}$

if a and b have the same remainder when they are divided by a non zero integer m .

Ex. 17 and 12 both give remainder 2 when divided by 5

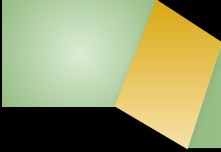
therefore $17 \equiv 12 \pmod{5}$



Congruence

If the remainders are different, then a and b are said to be not congruent modulo m , denoted $a \not\equiv b \pmod{m}$.

Ex. 7 and 5 give different remainder when divided by 3 therefore $7 \not\equiv 5 \pmod{3}$



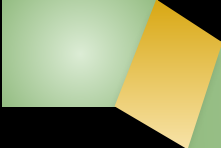
Congruence

By the definition of congruence, the following four equivalent relations are obvious :

$$a \equiv b \pmod{m} \Leftrightarrow a - b = km \Leftrightarrow a - b \equiv 0 \pmod{m} \Leftrightarrow m \mid (a - b)$$

Properties of Congruence

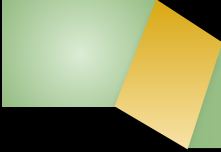
1. If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.
2. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
$$(a + c) \equiv (b + d) \pmod{m},$$
$$(a - c) \equiv (b - d) \pmod{m}$$



Properties of Congruence

3. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$.

4. If $a \equiv b \pmod{m}$ then $a^n \equiv b^n \pmod{m}$ for all natural numbers n .



Properties of Congruence

5. If $ac \equiv bc \pmod{m}$ and $(c, m) = 1$ then $a \equiv b \pmod{m}$.

Properties of Congruence

- ❖ If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.
- ❖ If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
$$(a + c) \equiv (b + d) \pmod{m},$$
$$(a - c) \equiv (b - d) \pmod{m}$$
- ❖ If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
$$ac \equiv bd \pmod{m}.$$
- ❖ If $a \equiv b \pmod{m}$ then $a^n \equiv b^n \pmod{m}$ for all natural numbers n .
- ❖ If $ac \equiv bc \pmod{m}$ and $(c, m) = 1$ then $a \equiv b \pmod{m}$.

Q. Reduce 8^{100} modulo 7.

Q. Reduce 8^{100} modulo 7.

Solution

Since $8 - (1) = 7$,

We can say that $8 \equiv 1 \pmod{7}$

Raising this congruence to the power of 100,

We have $8^{100} \equiv (1)^{100} \pmod{7}$

$$\Rightarrow 8^{100} \equiv 1 \pmod{7}$$

Q. Find the remainder when 3^{99} is divided by 13.

Q. Find the remainder when 3^{99} is divided by 13.

Solution

Since $27 - 1 = 26$,

We can say that $27 \equiv 1 \pmod{13} \Rightarrow 3^3 \equiv 1 \pmod{13}$

Raising this congruence to the power of 33,

we have $3^{99} \equiv (1)^{33} \pmod{13}$

$\Rightarrow 3^{99} \equiv 1 \pmod{13}$

Thus, the remainder is 1.

Q. Find the remainder of 7^{777} when it is divided by 10.

Q. Find the remainder of 7^{777} when it is divided by 10.

Solution

$$7^2 = 49 \equiv -1 \pmod{10}$$

$$\begin{aligned}\Rightarrow 7^{777} &\equiv (7^2)^{388} \cdot 7 \\ &\equiv 7 \pmod{10}\end{aligned}$$

Thus, the remainder is 7.

Note: For finding the remainder of a large power of a positive integer, it is important to find the minimum power with remainder 1, or see if the remainders are constant as the power changes.

Q. Prove that

(i) $47^{101} + 19^{101}$ is divisible by 66

(ii) $a^n - b^n$ is divisible by $a - b$, if n is odd

Q. Prove that

(i) $47^{101} + 19^{101}$ is divisible by 66

(ii) $a^n - b^n$ is divisible by $a - b$, if n is odd

Solution

$$(i) 47 \equiv -19 \pmod{66}$$

$$47^{101} \equiv (-19)^{101} \pmod{66}$$

Adding 19^{101} both sides,

$$47^{101} + 19^{101} \equiv (-19)^{101} + 19^{101} \pmod{66}$$

$$47^{101} + 19^{101} \equiv 0 \pmod{66}$$

$$(ii) a \equiv b \pmod{a - b}$$

$$a^n \equiv b^n \pmod{a - b}$$

$$a^n - b^n \equiv 0 \pmod{a - b}$$

Note:

$a^n + b^n$ is divisible by $(a + b)$ if n is odd

Q. Find the smallest positive integer k such that $9^{62} + k$ is divisible by 242.

A

120

B

142

C

161

D

201

Q. Find the smallest positive integer k such that $9^{62} + k$ is divisible by 242.

A

120

B

142

C

161

D

201

Q. Find the smallest positive integer k such that $9^{62} + k$ is divisible by 242.

Solution

$$3^5 \equiv 1 \pmod{242}$$

$$\Rightarrow 3^{5m} = (3^5)^m \equiv (1)^m \pmod{242} \equiv 1 \pmod{242}$$

$$\therefore (3^5)^{24}(3^4) \equiv 3^4 \pmod{242}$$

$$9^{62} \equiv 81 \pmod{242}$$

Therefore, the minimum value of k is equal to

$$242 - 81 = 161.$$

Q. Find the remainder when $452^{72^{452}}$ is divided by 3.

Q. Find the remainder when $452^{72^{452}}$ is divided by 3.

Solution

$452 \equiv -1 \pmod{3}$, thus

$$452^{72^{452}} \equiv (-1)^{72^{452}} \pmod{3} \equiv 1 \pmod{3}$$

The last congruence holds because 72^{452} is surely even.

Q. Find the last two digits of $(56789)^{41}$

A

39

B

59

C

89

D

99

Q. Find the last two digits of $(56789)^{41}$

A

39

B

59

C

89

D

99

Solution

$$56789 = 89(\text{mod } 100)$$

$$\equiv -11 (\text{mod } 100)$$

$$\therefore (56789)^{41} \equiv (-11)^{41} (\text{mod } 100)$$

$$\equiv (-11)^{40} \times (-11) (\text{mod } 100)$$

$$11^2 \equiv 21 (\text{mod } 100)$$

$$11^4 \equiv 41 (\text{mod } 100)$$

$$11^6 \equiv 21 \times 41 (\text{mod } 100)$$

$$\equiv 61 (\text{mod } 100)$$

$$11^{10} \equiv 41 \times 61 (\text{mod } 100)$$

$$\equiv 01 (\text{mod } 100)$$

$$11^{40} \equiv (01)^4 (\text{mod } 100)$$

$$\equiv 1 (\text{mod } 100)$$

$$(-11)^{41} \equiv 11^{40} \times (-11) (\text{mod } 100)$$

$$\equiv 1 \times (-11) (\text{mod } 100)$$

$$\equiv -11 (\text{mod } 100)$$

$$(56789)^{41} \equiv 89 (\text{mod } 100)$$

i.e, the last two digits of $(56789)^{41}$ is 89.

Q. The cube of any integer is of the form

A

$9k$

B

$9k + 1$

C

$9k + 8$

D

All of these

Q. The cube of any integer is of the form

A

$9k$

B

$9k + 1$

C

$9k + 8$

D

All of these

Q. The cube of any integer is of the form

Solution

$$n \equiv 0, \pm 1, \pm 2, \pm 3, \pm 4 \pmod{9}$$

$$n^3 \equiv 0, \pm 1, \pm 8, \pm 27, \pm 64 \pmod{9}$$

$$n^3 \equiv 0, \pm 1 \pmod{9}$$

$$= 9k, 9k+1, 9k+8$$

Q. Prove that $1^{29} + 2^{29} + \dots + (28)^{29}$ is divisible by 29.

Q. Prove that $1^{29} + 2^{29} + \dots + (28)^{29}$ is divisible by 29.

Solution

$$n - r \equiv -r \pmod{n}$$

$$\Rightarrow (n - r)^n \equiv (-r)^n \pmod{n}$$

$$\Rightarrow (n - r)^n + r^n \equiv 0 \pmod{n}, \text{ if } n \text{ is odd}$$

$$\therefore 1^n + (n - 1)^n \equiv 0 \pmod{n}$$

$$2^n + (n - 2)^n \equiv 0 \pmod{n}$$

$$\left(\frac{n-1}{2}\right)^n + \left(\frac{n+1}{2}\right)^n \equiv 0 \pmod{n}$$

Add these we get, $1^n + 2^n + \dots + (n - 1)^n \equiv 0 \pmod{n}$

Therefore, for our case, $n = 29$ which is odd.

Hence Divisible by 29

Q. Let $N = 1^{2017} + 2^{2017} + \dots + 2016^{2017}$. Then N is divisible by

A

2 but not 2017

B

2017 but not 2

C

Neither 2 nor 2017

D

Both 2 and 2017

Q. Let $N = 1^{2017} + 2^{2017} + \dots + 2016^{2017}$. Then N is divisible by

A

2 but not 2017

B

2017 but not 2

C

Neither 2 nor 2017

D

Both 2 and 2017

Q. Let $N = 1^{2017} + 2^{2017} + \dots + 2016^{2017}$. Then N is divisible by

Solution

By using the result $1^n + 2^n + \dots + (n - 1)^n$ is divisible by n for odd n , N is divisible by 2017.

Also, $1^{2017}, 3^{2017}, \dots, 2015^{2017}$ are all odd numbers which are 1008 in numbers, so, their sum is even.

Hence, N is divisible by 2 also.

Q. If $(2021)^{3762}$ is divided by 17, then the remainder is ____.

Solution

$$(2023 - 2)^{3762} = 2023k_1 + 2^{3762}$$

$$17k_2 + 2^{3762} \text{ (as } 2023 = 17 \times 17 \times 9 \text{)}$$

$$= 17k_2 + 4 \times 16^{940}$$

$$= 17k_2 + 4 \times (17 - 1)^{940}$$

$$= 17k_2 + 4 (17k_3 + 1)$$

$$= 17k + 4 \Rightarrow \text{remainder} = 4$$

Q. If the remainder when x is divided by 4 is 3, then the remainder when $(2020 + x)^{2022}$ is divided by 4 is

Solution

$$\text{Let } x = 4k + 3$$

$$(2020 + x)^{2022}$$

$$= (2020 + 4k + 3)^{2022}$$

$$= (4(505) + 4k + 3)^{2022}$$

$$= (4P + 3)^{2022}$$

$$= (4P + 4 - 1)^{2022}$$

$$= (4A - 1)^{2022}$$

Remainder is 1

Q. A five digit number $n = \overline{abcde}$ is such that when divided respectively by 2,3,4,5,6 the remainders are a, b, c, d, e. What is the remainder when n is divided by 100 ?

Q. A five digit number $n = \overline{abcde}$ is such that when divided respectively by 2,3,4,5,6 the remainders are a, b, c, d, e. What is the remainder when n is divided by 100 ?

Solution

$$n = \overline{abcde}$$

$$n \equiv a \pmod{2} \equiv b \pmod{3} \equiv c \pmod{4} \equiv d \pmod{5} \equiv e \pmod{6}$$

Possible remainders mod 2 = 0 or 1

But $a \neq 0$, so $a = 1$

Now since $n \not\equiv 0 \pmod{2}$, so its units digit must be 1, 3, 5, 7 or 9

So $e \in \{1, 3, 5, 7, 9\}$ but since 'e' is the remainder obtained on dividing n by 6, so it will be less than 6

$$\text{So } e \in \{1, 3, 5\}$$

$$\text{So } n = \overline{1bcd1}, \overline{1bcd3}, \overline{1bcd5}$$

Now, d is the remainder mod 5, so it will be 1, 3, 0 respectively

$$\text{So } n = \overline{1bc11}, \overline{1bc33}, \overline{1bc05}$$

Now, c is the remainder mod 4 and any number mod 4 is equivalent to last two digit mod 4.

So it will be 3, 1 and 1 respectively.

Solution

So $n = \overline{1_311}, \overline{1_133}, \overline{1_105}$

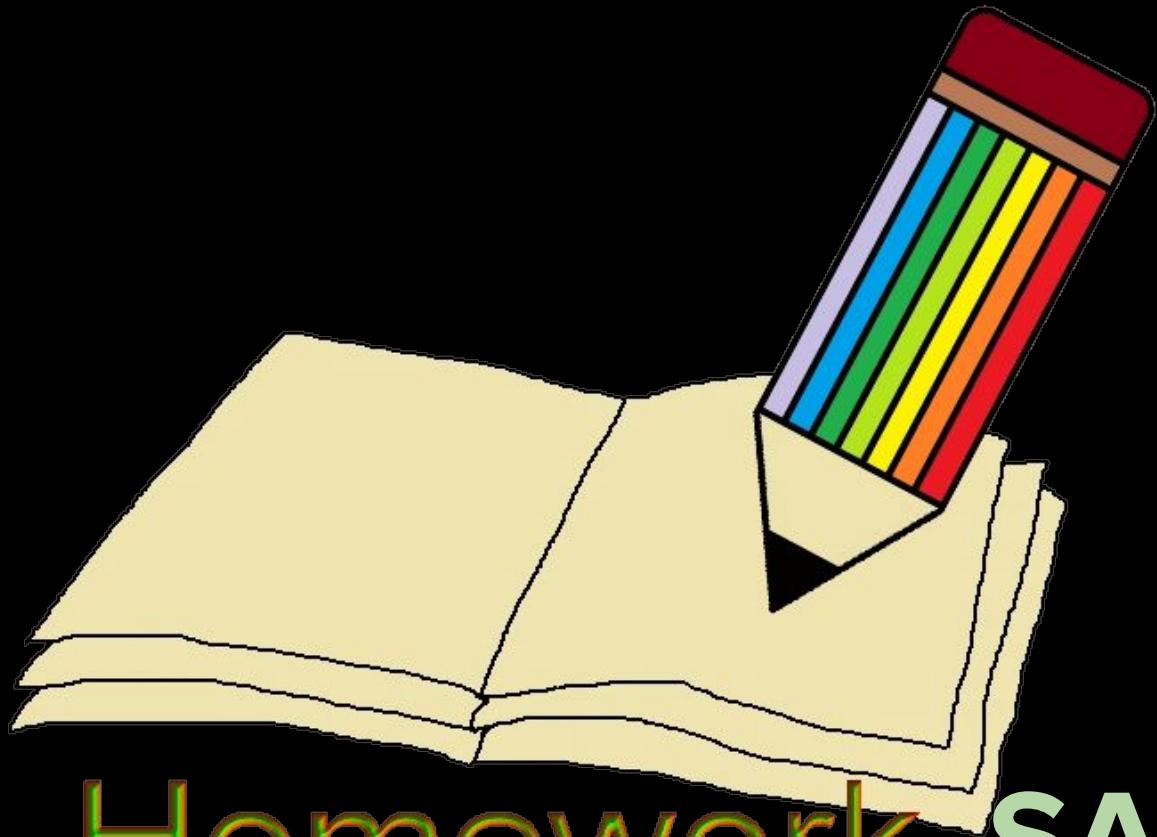
Now this vacant space is $b \pmod{3}$, which can be 0, 1 or 2

If $n = \overline{1_311}$	$n = \overline{1_133}$	$n = \overline{1_105}$
10311^x	10133^x	10105^x
$11311^{\checkmark}$	11311^x	11105^x
12311^x	12311^x	12105^x

So the number is $n = 11311$

and $11311 \equiv 11 \pmod{100}$

$\therefore \text{Ans} = 11$



Homework **SAWAAL**

Q. Let n be a number that is made from a string of 5s and is divisible by 2003. What is the last 6 digits of quotient when n is divided by 2003?

Note: Solution at the End



Homework **SAWAAL**

Q. Prove that for any natural number, n ,
 $E = 2903^n - 803^n - 464^n + 261^n$ is divisible by 1897.

Note: Solution at the End



NOTE: Solution also attached

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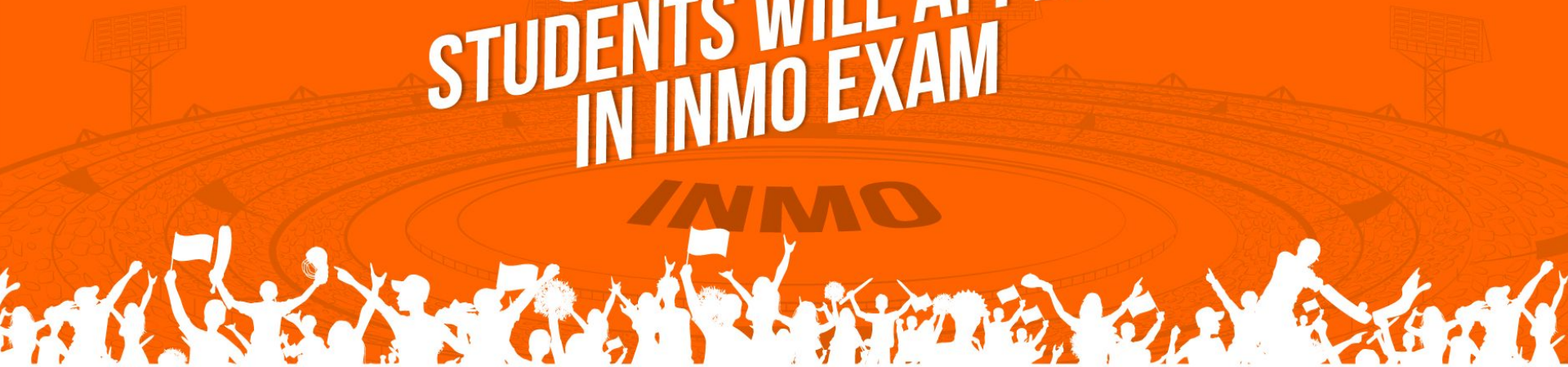
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Q. Let n be a number that is made from a string of 5s and is divisible by 2003. What is the last 6 digits of quotient when n is divided by 2003?

Solution

$$\text{Let } 2003x = 55\dots55555555$$

$$\Rightarrow 3x \equiv 555 \pmod{1000}$$

$$\Rightarrow x \equiv 185 \pmod{1000}$$

$$\Rightarrow x = 10^3y + 185 \text{ Say}$$

$$\Rightarrow 2003(10^3y + 185) = 55\dots55555555$$

$$\Rightarrow (2003000)y + 370555 = 55\dots55555555$$

$$\Rightarrow (2003000)y = 55\dots55185000$$

$$\Rightarrow 3y \equiv 185 \pmod{1000}$$

$$\Rightarrow 3y \equiv 1185 \pmod{1000}$$

$$\Rightarrow y \equiv 395 \pmod{1000}$$

$$\text{Hence, } x \equiv 395185 \pmod{1000000}$$

Q. Prove that for any natural number, n ,
 $E = 2903^n - 803^n - 464^n + 261^n$ is divisible by 1897.

Solution

$$1897 = 7 \times 271$$

$$\text{Now } (2903^n - 803^n) - (464^n - 216^n)$$

$$\text{As } (2903 - 803) \mid (2903^n - 803^n) \text{ and } (464 - 216) \mid (464^n - 216^n)$$

$$\text{i.e., } 2100 \mid (2903^n - 803^n) \text{ and } 203 \mid (464^n - 216^n)$$

$$\Rightarrow 7 \mid (2903^n - 803^n) \text{ and } 7 \mid (464^n - 216^n) \text{ (}\because 2100 = 7 \times 300 \text{ and } 203 = 7 \times 29\text{)}$$

Hence, $7 \mid E$

$$\text{Again, } 2903^n - 803^n - 464^n + 216^n = (2903^n - 464^n) - (803^n - 216^n)$$

$$2903 - 464 = 2439 \mid (2903^n - 464^n) \quad \text{and } (803 - 216) = 542 \mid (803^n - 216^n)$$

$$\text{i.e., } 2439 = 271 \times 9 \mid (2903^n - 464^n) \text{ and } 542 = 271 \times 2 \mid (803^n - 216^n)$$

So, $271 \mid (2903^n - 464^n)$ and $271 \mid (803^n - 216^n)$ and hence, $271 \mid E$.

Thus, the given expression is Divisible by the prime number 7 and 271

Hence, E is divisible by $271 \times 7 = 1897$